EXPLAINABLE DEEP LEARNING FOR PNEUMONIA DETECTION IN CHEST RADIOGRAPHS

Critical Reproducibility Analysis, Bias Mitigation, and Spatial Validation

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Numerical analysis for machine learning
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THE NEED FOR EXPLAINABLE AI (XAI) IN MEDICAL IMAGING



The Black-Box Problem: CNNs achieve high statistical accuracy in medical imaging, but lack diagnostic transparency.



Clinical Trust: Medical professionals cannot rely solely on binary outputs; they require spatial validation of the network's reasoning.



Our Objective: Implement, rigorously test, and scientifically validate the spatial interpretability of a ResNet-50 architecture for pneumonia detection.



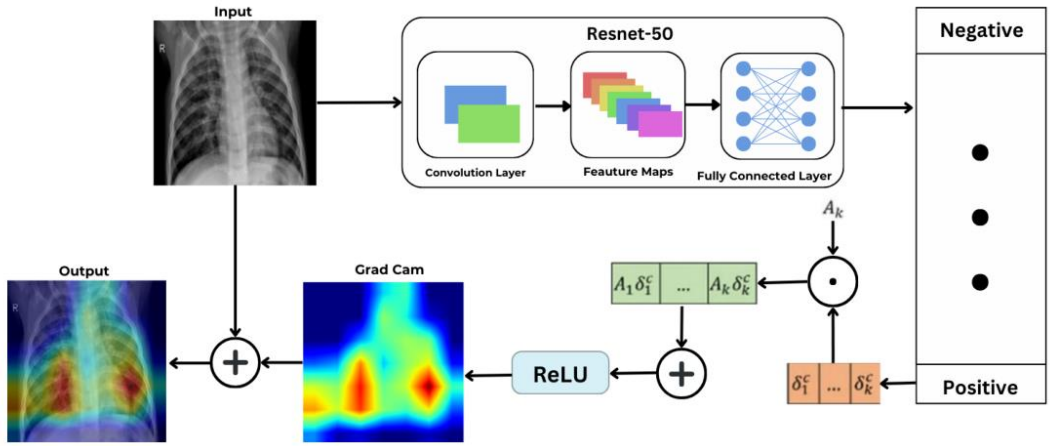


Starting Point: The Reference Paper

- **Reference:** *Interpretable Deep Learning for Pneumonia Detection Using Chest X-Ray Images* (Colin & Surantha, 2025).
- **The Challenge:** CNNs offer high diagnostic accuracy but act as "black boxes," limiting clinical trust. Additionally, medical datasets often suffer from severe class imbalance.
- **The Approach:** Mitigate class imbalance using data augmentation, while applying Explainable AI (XAI) techniques to visualize spatial focus and prove the model is looking at clinically relevant areas.
- **Key Finding:** Out of four evaluated XAI methods, Layer-wise Relevance Propagation (LRP) achieved the best interpretability (MRS = 0.85) and high accuracy (91%) with zero performance trade-off.

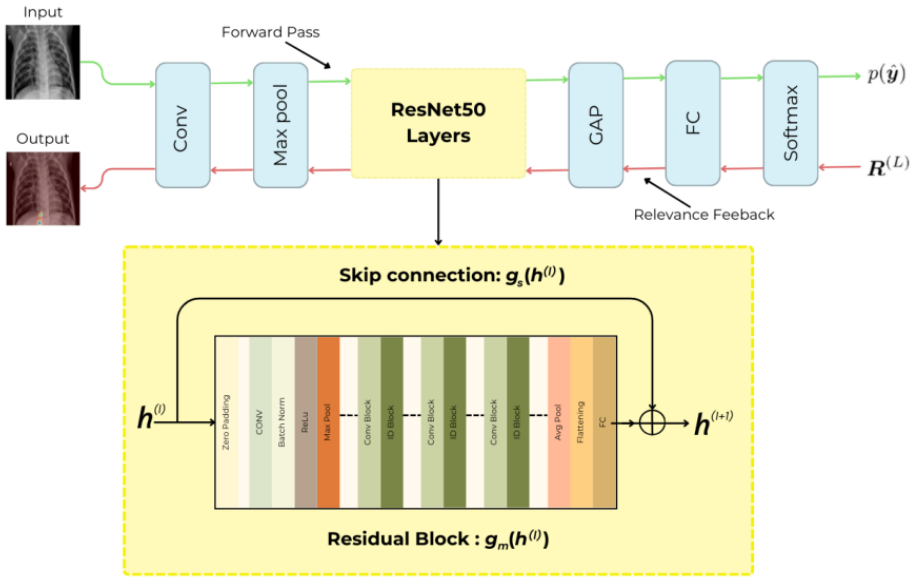
Explainability Methods: Grad-CAM

- Gradient-weighted Class Activation Mapping (Grad-CAM):
- Computes the gradient of the target class score with respect to the final convolutional feature maps.
- Generates a coarse localization map highlighting the regions most responsible for the prediction.
- Advantage: Highly efficient, post-hoc, and natively supported by PyTorch via the Captum library.



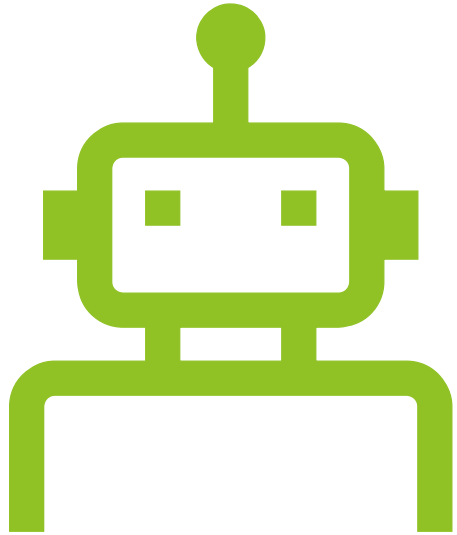
Explainability Methods: LRP

- Layer-wise Relevance Propagation (LRP):
- Operates on a strict conservation principle.
- Backpropagates a relevance score from the output class down to the individual input pixels.
- Assigns a strict mathematical contribution score to every pixel without relying on gradients.



Why We Focused Strictly on Grad-CAM

- The LRP Bottleneck on ResNet-50: LRP requires custom propagation rules for every mathematical operation.
- Skip Connections: Propagating relevance backward through ResNet's residual additions without violating the conservation property is mathematically unstable and highly intrusive.
- Our Decision: We anchored our pipeline to Grad-CAM. It provides robust, clinically validated spatial heatmaps without destabilizing the pre-trained ImageNet weights.



Methodological Flaws: Transfer Learning

The Inconsistency: The authors claim to use a pre-trained ResNet-50 with frozen upper layers.

The Reality: Their reported training trajectory starts at very low accuracy and climbs over 50 epochs.

Conclusion: This behavior is characteristic of training from scratch, contradicting the stated transfer learning strategy.

Methodological Flaws: The Post-Hoc Paradox

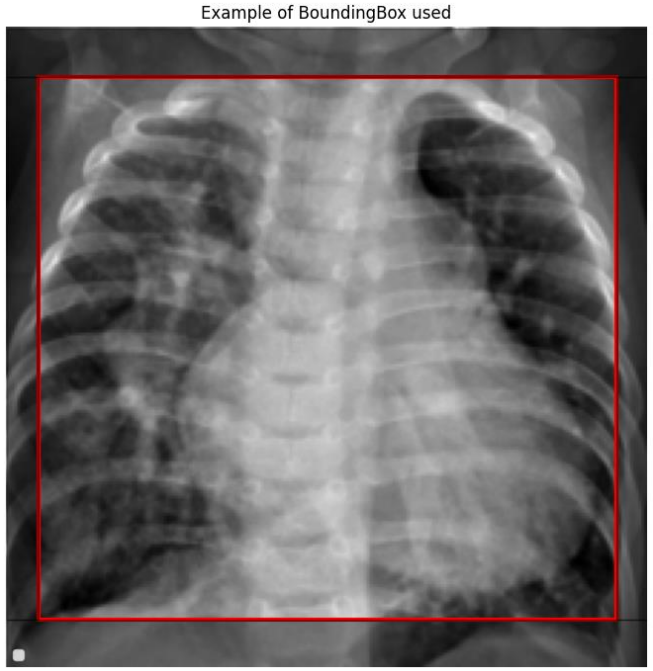
The Paradox: XAI techniques (LRP, Grad-CAM) are post-hoc; they explain a fixed model without altering its weights.

The Flaw: The reference paper reports varying baseline accuracies for different XAI methods (Base: 0.90, LRP: 0.91, CAM: 0.905).

Conclusion: Separate training runs were conducted for each technique, conflating standard run-to-run variance with the actual efficacy of the interpretability methods.

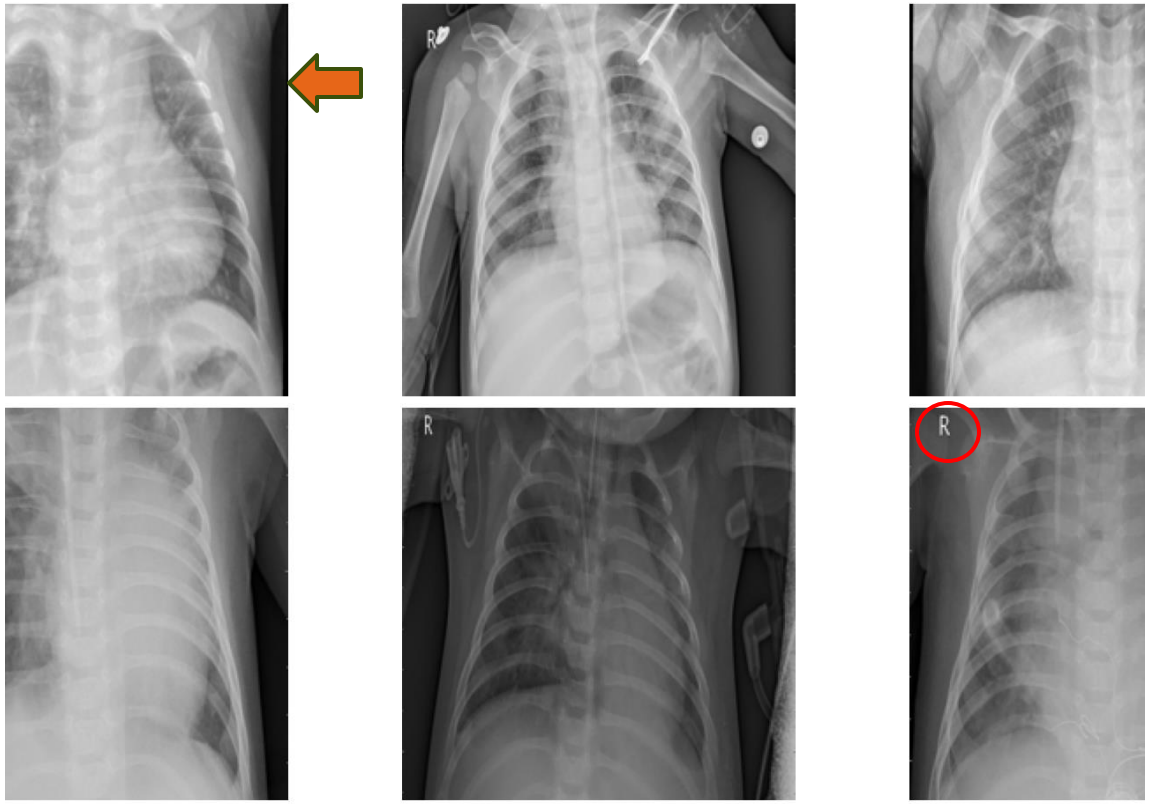
Methodological Flaws: The Spatial Ground Truth Dilemma

- The MRS Claim: The authors report a highly accurate Mean Relevance Score (MRS) of 0.85.
- The Dilemma: They used the Kaggle Guangzhou pediatric dataset, which contains zero spatial annotations or bounding boxes.
- Conclusion: It is mathematically impossible to compute a valid spatial alignment score (MRS) on a dataset lacking spatial ground truth.



$$\mathrm{MRS} = \frac{1}{N}\Sigma_{i=1}^{N} R_i$$

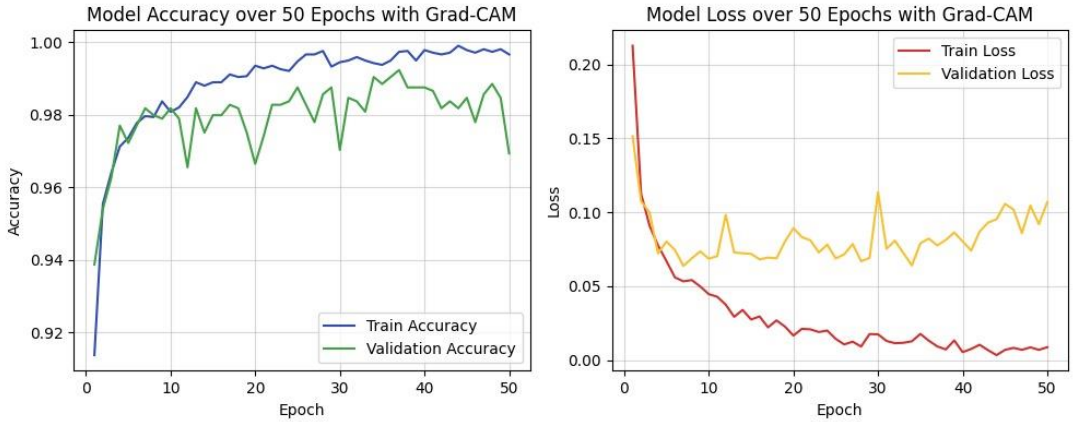
Initial Experiments & Geometric Bias Mitigation

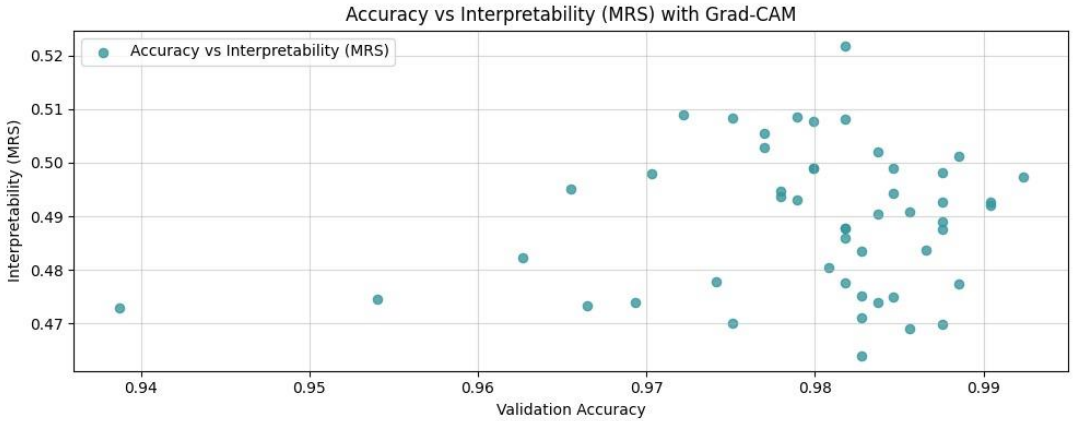


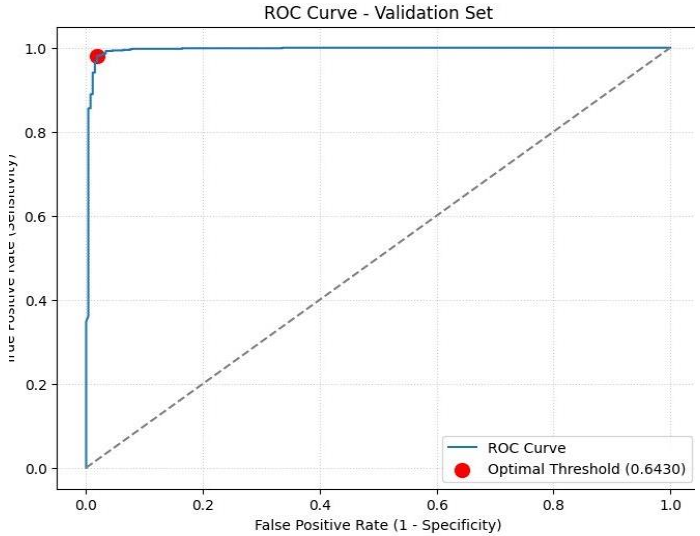
- Dataset: Kaggle Pediatric Chest X-rays.
- The Problem: Raw images contain severe peripheral biases (e.g., L/R lead markers).
- Mechanical Mitigation: Applied Resize(256) + CenterCrop(224) to mechanically excise borders.
- Static ROI: Defined a fixed central Region of Interest to temporarily bypass the lack of bounding boxes.

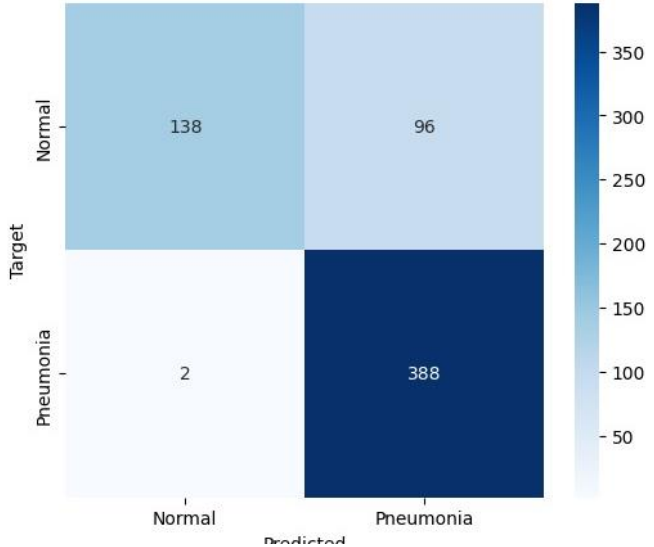
Phase 1: Training Dynamics & Classification

- Training: Stable convergence over 50 epochs using transfer learning.
- Optimal Threshold: Identified via Youden Index at 0.6430.
- Overall Performance: Accuracy of 84.29% and an impressive AUC-ROC of 0.9566.







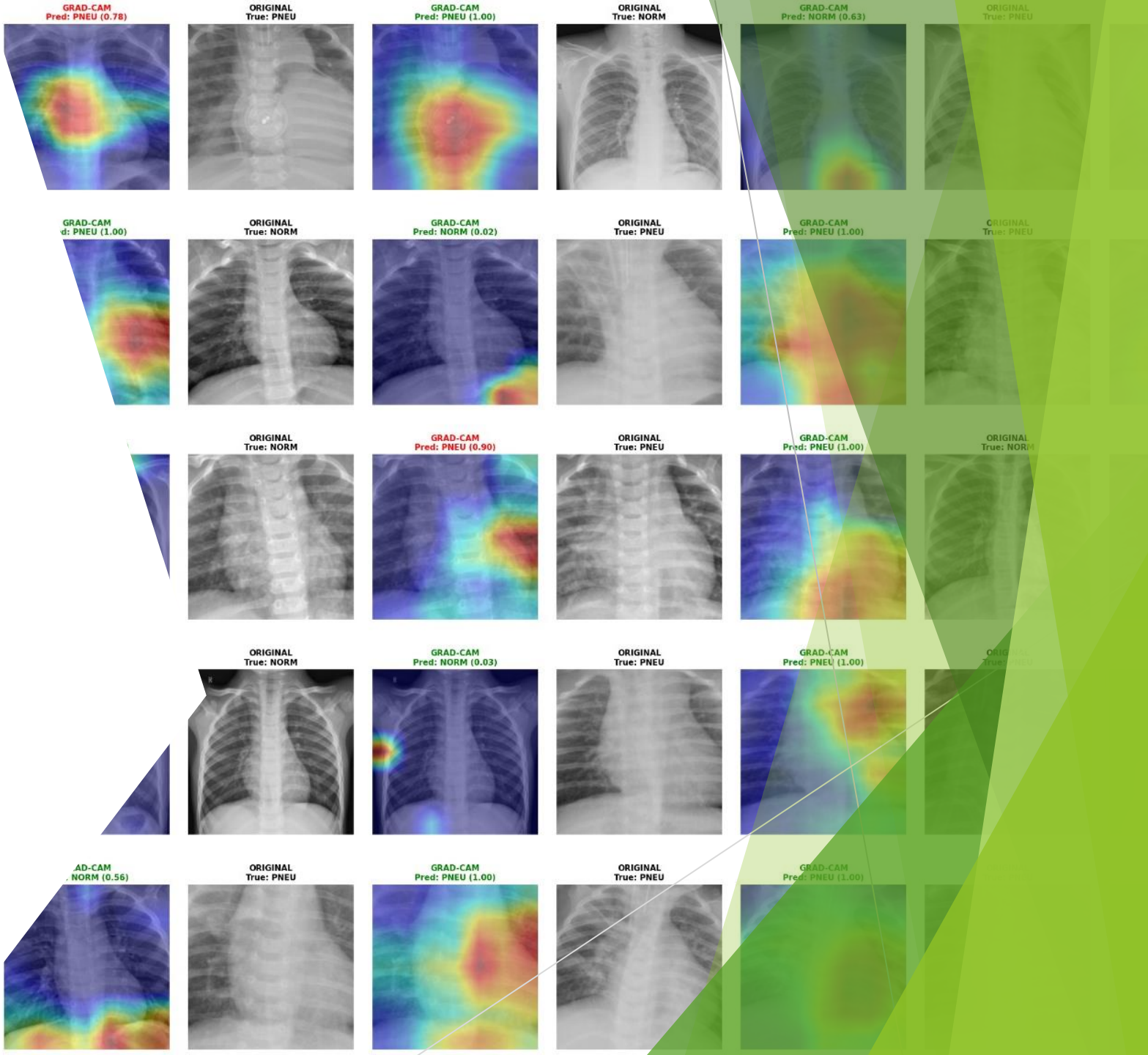


Phase 1: The Clinical Trade-off

- Sensitivity vs. Specificity: Near-perfect Sensitivity (0.99) but poor Specificity (0.58). The model is a conservative screener.
- XAI Validation: Grad-CAM successfully focuses on the central pulmonary regions, proving our geometric cropping strategy defeated peripheral shortcut learning.

Limitations of the Static ROI Strategy

- The Geometric Flaw: Assuming pneumonia always occurs centrally is clinically incorrect. A static ROI penalizes true peripheral opacities and rewards central noise.
- The Scientific Conclusion: Claiming an accurate MRS requires true clinical annotations.

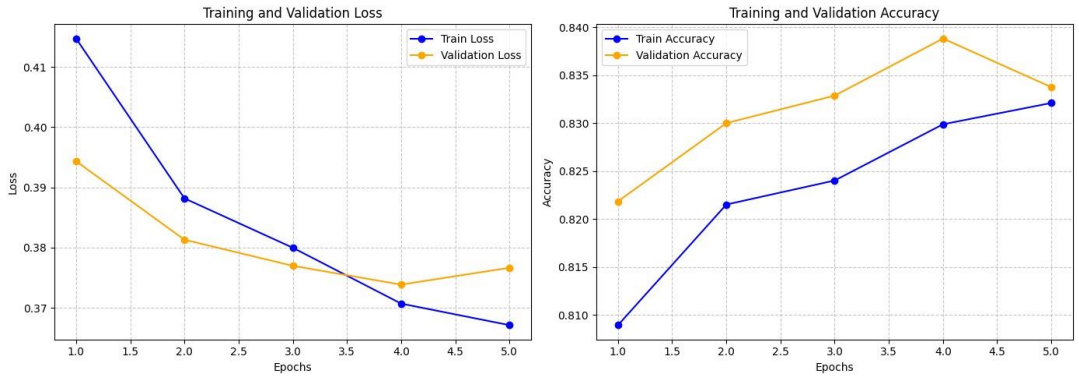


The Dataset Pivot: RSNA Adult Benchmark

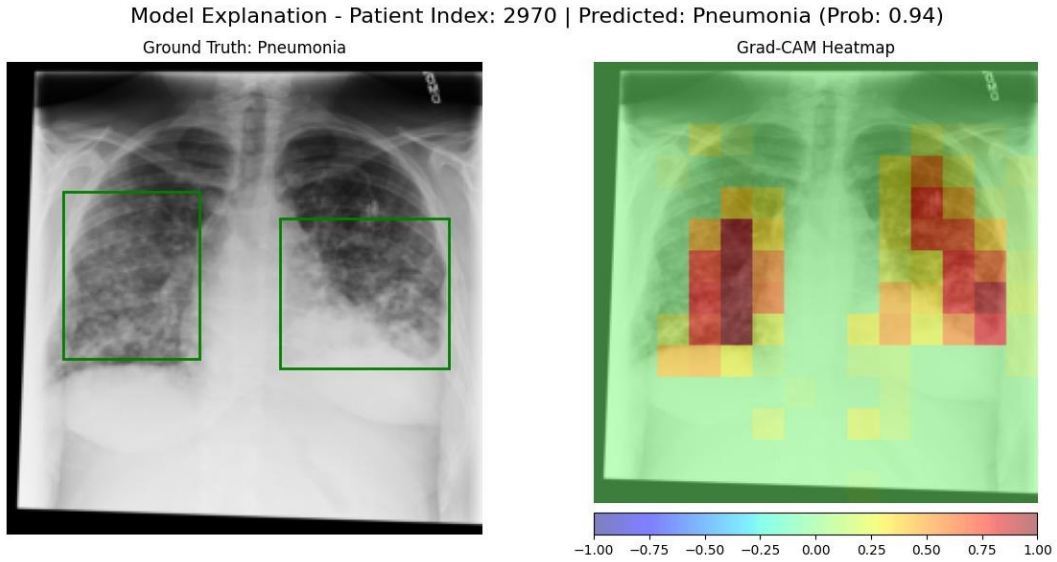
- The Upgrade: ~30,000 adult chest radiographs.
- The Key Feature: Includes expert-annotated ground-truth bounding boxes for all positive opacities.
- Clinical Complexity: Adult opacities are notoriously subtle and comorbid, presenting a strictly harder baseline.
- Mathematical Rigor: We implemented dynamic coordinate scaling to programmatically compute the true Relevance Mass Ratio against exact bounding boxes.

Training on the RSNA Dataset

- Targeted Unfreezing: Unfroze layer3 and layer4 (learning rate = 0.001) to capture complex adult pulmonary textures.
- Rapid Convergence: Optimal adaptation achieved rapidly due to dataset volume.
- Early Stopping: Training halted strictly at 5 epochs as validation loss began to plateau/rise, preventing early overfitting.



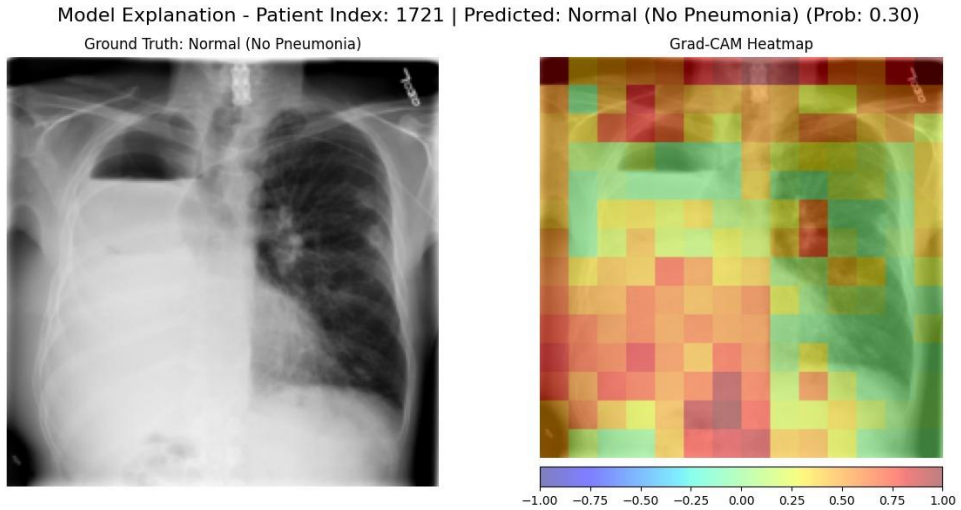
Objective Spatial Validation: Positive Case



- Patient Index 2970 (Probability: 0.94).
- Observation: The network's high-relevance activations perfectly align with the clinical ground-truth bounding boxes.
- Conclusion: The model actively detects and localizes true bilateral infiltrates rather than relying on global image artifacts.

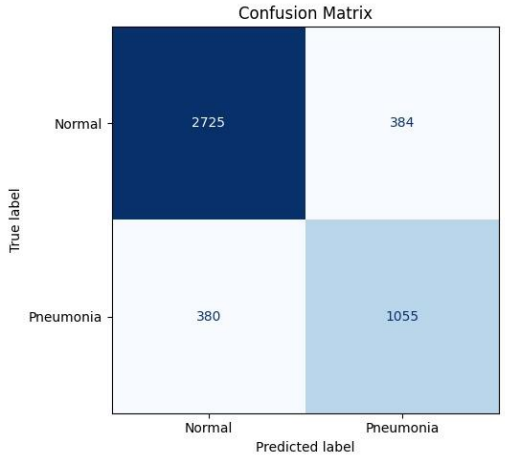
Objective Spatial Validation: Negative Case & Global MRS

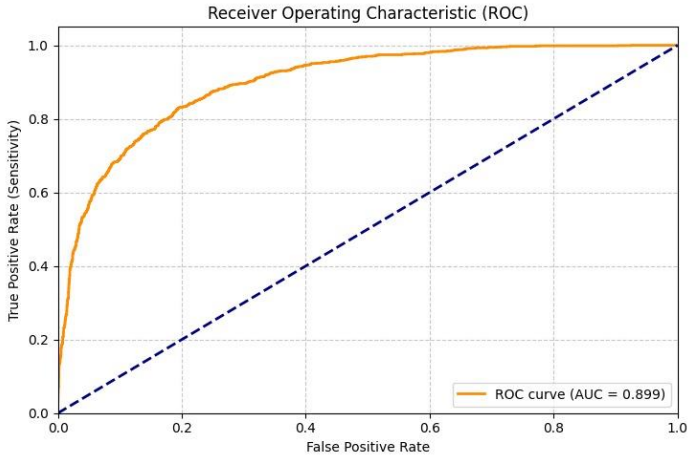
- Patient Index 1721 (Probability: 0.30).
- Observation: The negative ground truth presents no bounding boxes. The Grad-CAM correctly diffuses its activations, avoiding false localized alerts.
- Global MRS: Evaluated over 894 unique positive test images, the model achieved a Global MRS of 0.3013, mathematically certifying robust spatial alignment.



RSNA Clinical Evaluation Metrics

- Robust Separation: AUC-ROC of 0.899.
- Accuracy: Final test accuracy of 83.19% on 4,544 unseen patients.
- Balanced Screening: Specificity (Normal): 87.6% (Precision 0.88). Sensitivity (Pneumonia): 73.5% (Precision 0.73).
- Conclusion: A highly reliable and balanced screening tool for complex adult radiographs.





Final Conclusions



Reproducibility: Demonstrated that XAI claims based on datasets without spatial annotations are scientifically unverifiable.



Methodological Rigor: Successfully implemented a rigorous XAI pipeline combining Grad-CAM, dynamic scaling, and exact MRS formulation.



Clinical Impact: Proved that transfer learning coupled with Explainable AI can yield diagnostic tools that are both statistically accurate and transparently interpretable for clinicians.